

3 Elimination of variables.

(4)

3.1 Elimination ideals.

3.1.1 Def. Let $I \subseteq k[x_1, \dots, x_n]$ be an ideal and $\ell \in \{0, \dots, n\}$. Then $I_\ell := I \cap k[x_{\ell+1}, \dots, x_n]$ is called the ℓ -th elimination ideal of I .

3.1.2 Proposition. The ℓ -th elimination ideal I_ℓ of an ideal $I \subseteq k[x_1, \dots, x_n]$ is an ideal in the ring $k[x_{\ell+1}, \dots, x_n]$.

Proof: $0 \in k[x_{\ell+1}, \dots, x_n]$ and $0 \in I$, so $0 \in I_\ell$. Because every ring is closed under $+$ and every ideal, too,

we see: $f, g \in I_\ell \Rightarrow f+g \in I_\ell$.

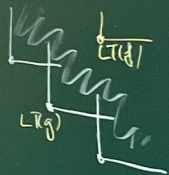
The last property to check: $f \in k[x_{\ell+1}, \dots, x_n]$ and $g \in I_\ell = I \cap k[x_{\ell+1}, \dots, x_n]$, then $f \cdot g \in k[x_{\ell+1}, \dots, x_n]$ and $f \cdot g \in I$, so $f \cdot g \in I_\ell$.

3.1.3 Remark. We will treat Gröbner bases as sets and we assume that $\langle \emptyset \rangle = \{0\}$ (empty set generates the zero ideal). $\square$

3.1.4 Theorem (Elimination theorem). Let $I \subseteq k[x_1, \dots, x_n]$ be an ideal and $\ell \in \{0, \dots, n\}$ and let G be a Gröbner basis of I with respect to the lex order. Assume $I \neq \{0\}$. Then $G \cap k[x_{\ell+1}, \dots, x_n] =: G_\ell$ is a Gröbner basis with respect to the lex order for the ℓ -th elimination ideal $I_\ell = I \cap k[x_{\ell+1}, \dots, x_n]$.

Proof. Just by construction, $G_\ell \subseteq I_\ell$. Pick any $f \in I_\ell \setminus \{0\}$. To show that G_ℓ is a (2)

Groebner basis, we need to show $LT(f) \in \langle LT(g) : g \in G_\ell \rangle$. G is a Groebner basis of I , so $LT(f) \in \langle LT(g) : g \in G \rangle$, but remember $f \in I_\ell$ so that



$f \in k[x_{\ell+1}, \dots, x_n]$. The exponent vector of $LT(f)$ is $\alpha = (0, \dots, 0, \alpha_{\ell+1}, \dots, \alpha_n)$.

So, since G is a Groebner basis, it contains some polynomial $g \in G$ with the leading term having the exponent vector β that satisfies $\beta \leq \alpha$. But since $\alpha = (0, \dots, 0, \alpha_{\ell+1}, \dots, \alpha_n)$, β , too, has the form $\beta = (0, \dots, 0, \beta_{\ell+1}, \dots, \beta_n)$.

But since order is a lex order and β is the exponent for the leading term, all other terms of g will have exponent vectors of the form $\gamma = (0, \dots, 0, \gamma_{\ell+1}, \dots, \gamma_n)$. $\Rightarrow g \in k[x_{\ell+1}, \dots, x_n]$.

So, $g \in G_\ell$. This shows, that G_ℓ is indeed a Groebner basis of I_ℓ with respect to the lex order. $\square$

$$\begin{pmatrix} 0, 0, 2, 3 \\ 0, 0, 1, 1000 \end{pmatrix}$$